



PROJECT REPORT

Submitted by,

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TOPIC

**CAMPUS NETWORK
MANAGEMENT ARCHITECTURE**

CERTIFICATE

This is to certify that this project report entitled "**CAMPUS NETWORKING**" is a bonafide record of the work done by BIJOY SIMON towards the partial fulfilment of the requirement for the reward of CCNA, under our supervision during the Period of 1/08/2025 - 27/08/2025.

PROJECT COORDINATOR

SHALINI

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AIM

- To start a networking system for a campus management architecture.
- To establish a communication between different departments and buildings.

PROCEDURES

First design the architecture.

- Insert the network devices and components.
- After that give IP address to all devices and make all connections (devices) up.
- Then start configuration with required commands.
- Apply the routing protocol RIP(Routing information protocol).
- Apply Extended Acl to restrict building a access to building C,D and E.
- Intervlan to communicate between different vlans
- Check whether the communication is established.
- Vtp connection to communicate between switches.

Abstract

A network system has to be developed for the campus management system. The campus consists of 5 blocks named A B C D and E . In block A has two department admin and finance. In block B has two department named restroom staffroom . The entire network using rip for communication. Block A and B are using intervlan communication. All blocks using dhcp service. Block E is a document holder that is all documents are stored in this block. So we want to set ssh configuration in this block for more security and enable AAA authentication. add and check atleast 6 username and password and also add routers. Enable ftp protocol in that server also and check it work well efficiently the entire network. All switches keep a vtp connection also Use HSRP IN BLOCK E Do backup and restore in all the blocks and verify its using another router .ACL working in block D in last moment .Connect any wireless architecture

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NETWORK REQUIREMENTS

- Identify the hardware components that are required for the networking of the campus management architecture.
- IP network designs for different departments.
- IP addressing range for every devices.
- The users at different departments or buildings should be able to access each other as per the architecture.

THEORY

Routing :

Routing is a method to find the best path between source to Destination.

Administrative distance (AD) :

Administrative distance is the default values assigned for routing protocols. It is used to determine which routing protocol is to trust most, when router may have more than one routing protocols. It will choose a Routing protocol having lowest AD.

Administrative Distance	Default Value
Static Routing	1
RIP	120
EIGRP	90
OSPF	110

Default Routing

- When a packet arrives on an interface of a router, the router checks the destination network address of the arrived-packet and finds that destination address in the routing table. If the router finds an entry for the destination network, the router forwards the incoming packet from the interface that is specified in the entry.
- If the router does not find an entry for the destination network, the router checks the default route. If the default route is available, the router forwards the incoming packet from the interface that is specified in the default route.
- If the default route is not available, the router drops the incoming packet. A router uses a default route to forward all incoming packets; those destination addresses are not available in the routing table.

Static Routing

- In this every path is set manually.
- The administrator have full control of network.
- It is more secure than other routing protocol.
- It have very low CPU usage.
- Routes are added manually in routing table.

- If one link fail, then data will not reach destination
- Static Routing Commands
- Routes are added manually in routing table.
- If one link fail, then data will not reach destination.

Static Routing Commands

- **To configure static routing**

Router(config)#iproute

Eg: Router(config)#ip route 200.0.0.0 255.255.255.0 10.0.0.1

- **To view routing table**

Router#show ip route static

- **Remove the route from routing table**

Eg: Router(config)#no ip route 200.0.0.0 255.255.255.0 10.0.0.1

Default Routing

- Default router allows packet to be forwarded even without a specific route to a particular network.
- Default router is identified by all zeros in both network and subnet mask

To configure Default Route

Router(config)#ip route 0.0.0.0 0.0.0.0

Classful Routing Protocol

Routing protocols do not send subnet mask information with their routing updates

Eg: RIPv1 and IGRP.

Classless Routing Protocol

Routing protocols send subnet mask information with their routing updates

Eg: RIPv2, EIGRP, OSPF, RIP (Routing Information Protocol)

RIP (Routing Information Protocol)

- Distance vector routing protocol

//RIP sends out periodic route updates every 30s

- Rip is designed for use on smaller networks.

//Rip has maximum hop count 15

//total hc = no. of Routers – 1

- RIP uses metric as Hop count
- RIP uses Bellman-Ford algorithm to find best path
- RIP has administrative distance of 120 for RIPv1

RIP v1 classful

- RIP v1 sends updates as broadcast to address using 255.255.255.255
- Hop count 15, Administrative distance 120 RIPV1 Configuration

Router(config)#router rip

Router(config-router)#network RIP V2

RIP v2 classless

- RIP 2 sends Routing updates via multicast, using address 224.0.0.9
- Hop count 15, Administrative distance 120 RIPV2 Configuration

Router(config)#router rip Router(config-router)# Version 2

Router(config-router)#network

EIGRP

- Enhanced interior gateway routing protocol
- Eigrp is a no cisco proprietary routing protocol
- Administrative distance 90
- Eigrp is a classless routing protocol

- Eigrp uses diffusing update algorithm (DUAL) to find best path.
- Configuring EIGRP

Router(config)#router eigrp

//same in all router

Router(config-router)#network Autonomous number must be same in all routers.(1-65535)

OSPF

- OSPF IS link-state routing protocol.
- Routing information is shared through link-state advertisement (LSAS).
- Routing information is shared through link-state advertisement.(LSAS)
- OSPF will form neighbor relationship with adjacent routers in same area
- OSPF have administrative distance 110

OSPF uses cost as its metric

- OSPF is a classless protocol OSPF configuration

Router(config)#router ospf

Router(config-router)#network area

Note: area number should be same in all routers.

VLAN

- Vlans(Virtual Lans)are logical grouping of ports in cisco switch.
- It consists of diff vlan. vlan 1,1002,1003,1004,1005
- By default all switchport will be the member of vlan 1 that's y switches have a single broadcast domain.
- In vlan we have a range of 2-1001 because vlan 1 and 1002 is already exist
- Each port of a switch can either be a access port or a trunk port
- Access port are ports used for vlan creation
- To configure an interface to be an access interface,the switchport mode access interface command is used.

VLAN configuration

- Change name of vlan

Switch(config)#vlan

Switch(config-vlan)#name

Switch(config-vlan)#exit

- Adding a port to vlan

Switch(config-if)#interface name id

Switch (config-if)switchport mode access

Switch (config-if)switchport access vlan

- Adding a group of ports to vlan

Switch (config-if)#interface
rangeinterfacename,interfacename

Switch (config-if)#switchport mode access

Switch (config-if)#switchport access vlan

- To view show vlan status

Switch#show vlan

Access control list (Acl) :

- A set of rules which will allow or deny the specific traffic moving through the route.
- It is a Layer 3 security which controls the flow of traffic from one router to another.
- It is also known as Packet filtering firewall.

Types of Acl:

- Standard (Numbered, Name)
- Extended (Numbered, Name)

Standard ACL: (Numbered)

- Acl number range 1-99
- We allow or deny a network, host and subnet.
- All services are blocked.
- Filtering is done based on only source IP address.
- Implemented closest to the destination.

Standard Acl Configuration

```
Router(config)# access-list(access-list number){permit/deny}  
host (IP address)
```

```
Router(config)# access-list (access-list number) permit any
```

Applying Acl to an interface:

```
Router(config)#interface interface name
```

```
Router(config-if)# ip access-group(access-list number {in/out}
```

```
Router(config-if)exit
```

Extended Acl: (Numbered)

- Acl number range is 100-199
- We can allow or deny a network, host, subnet and services.
- Selective services can be blocked.
- Filtering is done based on source IP , destination IP, protocol port number address.
- Implemented closet to the source.

Extended Acl configuration:

```
Router(config)#access-list <access-list num> {permit/deny}  
<protocol> host <source ip> host <destination ip> eq<port no>
```

```
Router(config)#access-list <access-list num> permit ip any any
```

Standard ACL: (Named)

- Acl with name are called Named Acl's

Configuration:

```
Router(config t)#ip access-list standard "access-list name"
```

```
Router(config t- std-nacl)#deny host <address>
```

```
Router(config t-std-nacl)#permit any
```

Extended Acl: (Named)

```
Router(config t)# ip access-list extended 'access-list name'
```

```
Router(config t -ext-nacl)#deny 'service' host 'address'
```

```
Router(config t-ext-nacl)#permit icmp any any
```

VTP:

- VTP means vlan trunking protocol
- We know trunk is used to transmit vlan information between switches.
- Vlan trunking protocol is a cisco proprietary protocol that propagate the definition of virtual local area network(vlan) on the whole local area network.
- Vtp carries vlan information to all the switches in vtp domain.
- Vtp can manually add/ remove /rename vlans.

Types of vtp:

VTP client mode:

- A switch using this mode cannot change its vlan configuration.
- A vtp client switch cannot create or delete vlans
- Received vtp updates are processed and forwarded.

VTP server mode:

- A switch using this mode can create and delete vlans.
- A vtp server switch will propagate vlan changes.
- This is the default mode for cisco switches.

VTP transparent mode:

- A switch using this mode doesn't share its vlan database.
- It forwards received vtp advertisements.
- You can create and delete a vlan on a vtp transparent switch, the changes are not sent to other switches.
- Changes done on vtp server don't affect vtp transparent.

//to add domain name

Switch(config)#vtp domain name

//vtp transparent mode configuration

Switch(config t)# vtp mode transparent

//vtp client mode configuration

Switch(config t)# vtp mode client

//to check vtp status

Switch# show vtp status

AAA (Authentication, Authorization, and Accounting)

In networking, AAA stands for Authentication, Authorization, and Accounting. It's a security framework that controls access to network resources and keeps track of user activity. AAA ensures that only authorized users can access the network, limits what they can do, and records their actions for auditing and billing purposes.

Here's a breakdown of each component:

Authentication:

Verifies the identity of a user or device trying to access the network. This often involves checking usernames and passwords, or using other authentication methods like certificates or multi-factor authentication.

Authorization:

Determines what a user or device is allowed to do once they have been authenticated. This might involve restricting access to certain files, applications, or network services.

Accounting:

Tracks user activity on the network, including login times, resources accessed, and actions performed. This information is valuable for auditing, troubleshooting, and billing.

AAA is a critical part of network security, helping to protect sensitive resources and ensure proper network usage, according to network security resources. It provides a structured approach to access control, enabling network administrators to manage and secure their networks effectively.

DHCP (Dynamic Host Configuration Protocol)

The dynamic host configuration protocol (dhcp) is a network management protocol used on internet protocol networks whereby a dhcp server dynamically assign an ip address and other network configuration parameters to each device on a network.so they can communicate with other ip network.

Dhcp relay agent:

A dhcp relay agent is a computer or router that listen for dhcp broadcast from dhcp clients and then relays (sends) those messages to dhcp servers on the another networks.Requesting dhcp services on router from server

```
Router(config)#int (int id)
```

```
Router(config)#ip helper-address (server ip)
```


FTP (File Transfer Protocol)

FTP is a standard network protocol used to transfer files between a client and a server over TCP/IP networks. It usually works on TCP port 21 (control connection) and port 20 (data connection)

FTP Service Features

Allows users to upload/download files to/from the server. Supports anonymous access (public) or authenticated access (username/password). Can operate in Active Mode or Passive Mode (for firewall/NAT compatibility). Supports permissions and directory restrictions for security

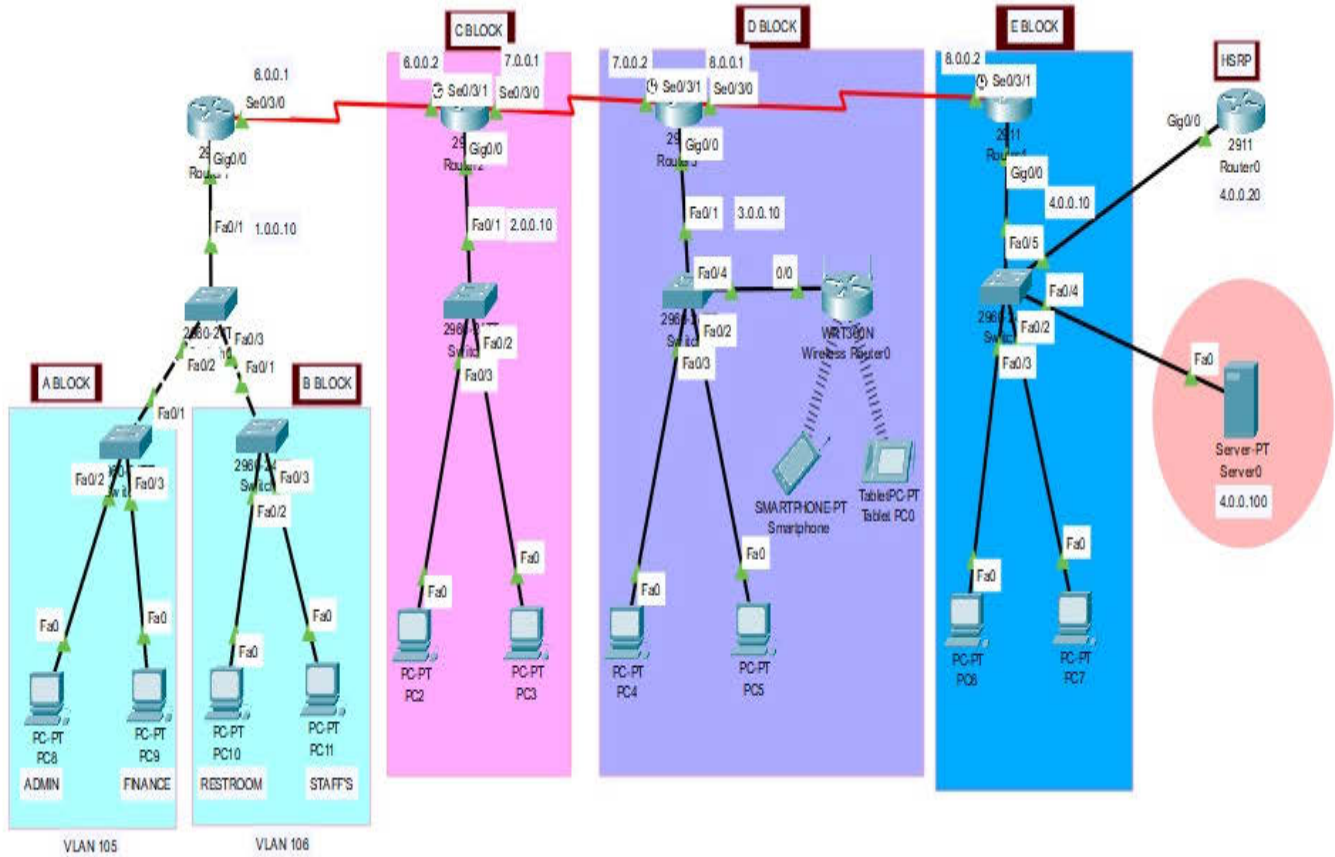
HSRP(Hot Standby Router Protocol)

Hot Standby Router Protocol (HSRP) is a CISCO proprietary protocol, which provides redundancy for a local subnet. In HSRP, two or more routers give an illusion of a virtual router. HSRP allows you to configure two or more routers as standby routers and only a single router as an active router at a time. All the routers in a single HSRP group share a single MAC address and IP address, which acts as a default gateway to the local network. The *Active router* is responsible for forwarding the traffic. If it fails, the *Standby router* takes up all the responsibilities of the active router and forwards the traffic.

Wireless Networking

A wireless network is a computer network that uses wireless data connections between network nodes. Wireless networking allows homes, telecommunications networks, and business installations to avoid the costly process of introducing cables into a building, or as a connection between various equipment locations. Admin telecommunications networks are generally implemented and administered using radio communication. This implementation takes place at the physical level (layer) of the OSI model network structure. Examples of wireless networks include cell phone networks, wireless local area networks (WLANs), wireless sensor networks, satellite communication networks, and terrestrial microwave networks.[4]

Topology



Configuration:

Building A B(Administrative block):

//up the connection in interface fa0/0 (Router 1 (AB))

Router1>en

Router1>config

Router1(config)#Interface gig0/0

Router1(config)#ip address 1.0.0.10 255.0.0.0

Router1(config -if)#No shutdown

Router(config -if)#Exit

//up the connection in Serial interface fa0/0 (Router 1

(AB)) Router1(config)#Interface Se0/0/0

Router1(config)#ip address 6.0.0.1 255.0.0.0

Router1(config -if)#No shutdown

Router(config -if)#Exit

//Create two departments vlan 105 and vlan 106.

Vlan 105 - Ablock Vlan 106 - Bblock

Switch0> en

Switch0> config t

Switch0(config)#vlan 105

Switch0(config)#name A

Switch0(config)#Exit

Switch0(config)#Vlan 106

Switch0(config)#Name B

Switch0(config)#Exit

Switch0(config)#Interface range fa0/2-3

Switch0(config -if)#Switchport mode access

Switch0(config -if)#Switchport access vlan 105

Switch0(config -if)#Exit

Switch0(config)#Interface range fa0/4-5

Switch0(config -if)#Switchport mode access

Switch0(config-if)#Switchport access vlan 106

Switch0(config -if)#Exit

// connect interface fa0/1 Trunk

```
Switch0(config )#Interface fa0/1  
Switch0(config -if)#Switchport mode trunk  
Switch0(config -if)#Exit
```

// connection between two different vlans by Intervlan

```
AB(config )#interface fa0/0.1  
Router1(config -if)#Encapsulation dot1q 105  
Router1(config -if)#Ip address 192.168.5.1 255.255.255.0  
Router1(config -if)#No shutdown  
Router1(config -if)#Exit  
Router1(config )#Interface fa0/0.2  
Router1(config -if)#Encapsulation dot1q 106  
Router1(config -if)#Ip address 192.168.6.1 255.255.255.0  
Router1(config -if)#No shutdown  
Router1(config-if)#Exit
```

VTP

// Main Switch in A,B Block

```
Switch(config-if)#int range fa0/2-3
```

```
Switch(config-if-range)#switchport mode trunk
```

```
Switch(config-if-range)#vtp mode server
```

Device mode already VTP SERVER.

```
Switch(config)#vtp domain ccna
```

Changing VTP domain name from NULL to ccna

// A Block Switch

```
Switch(config)#int fa0/1
```

```
Switch(config-if)#switchport mode trunk
```

```
Switch(config-if)#vtp mode client
```

Setting device to VTP CLIENT mode.

```
Switch(config)#vtp domain ccna
```

Domain name already set to ccna.

// B block Switch

Switch(config)#int fa0/1

Switch(config-if)#switchport mode trunk

Switch(config-if)#vtp mode client

Setting device to VTP CLIENT mode.

Switch(config)#vtp domain ccna

Domain name already set to ccna.

Block C

//up the connection (Router 2)

Router2>en

Router2#Config t

Router2(config)#Interface gig0/0

Router2(config-if)#ip address 2.0.0.10 255.0.0.0

Router2(config-if)#No shutdown

Router2(config-if)#Exit

//up the connection in Serial interface fa0/0 (Router 2)

```
Router2(config)#Interface Se0/0/0
```

```
Router2(config)#ip address 6.0.0.2 255.0.0.0
```

```
Router2(config-if)#No shutdown
```

```
Router2config-if)#Exit
```

```
Router2(config)#Interface Se0/0/1
```

```
Router2(config)#ip address 7.0.0.1 255.0.0.0
```

```
Router2(config-if)#No shutdown
```

```
Router2(config-if)#Exit
```

Block D

// up the connection in the interface fa0/0 (Router 3)

```
Router3>en
```

```
Router3#config t
```

```
Router3(config)#interface gig0/0
```

```
Router3(config-if)#ip address 3.0.0.10 255.0.0.0
```

```
Router3(config-if)#no shutdown
```

```
Router3(config-if)#exit
```

//up the connection in Serial interface (Router 3)

```
Router3(config)#Interface Se0/0/1
```

```
Router3(config)#ip address 7.0.0.2 255.0.0.0
```

```
Router3(config-if)#No shutdown
```

```
Router3config-if)#Exit
```

```
Router3(config)#Interface Se0/0/0
```

```
Router3(config)#ip address 8.0.0.1 255.0.0.0
```

```
Router3(config-if)#No shutdown
```

```
Router3(config-if)#Exit
```

Building E

//up the interface connection

```
Router4>en
```

```
Router4#config
```

```
Router4(config)#interface gig0/0
```

```
Router4(config-if)#ip address 4.0.0.10 255.0.0.0
```

```
Router4(config-if)#no shutdown
```

```
Router4(config-if)#Exit
```

//up the connection in Serial interface (Router 4)

Router2(config)#Interface Se0/0/0

Router2(config)#ip address 8.0.0.2 255.0.0.0

Router2(config-if)#No shutdown

Router2(config-if)#Exit

Routing

//Router1 routing protocol RIP

```
Router1(config)#router rip
```

```
Router1(config-rip)#network 1.0.0.0
```

```
Router1(config-rip)#network 6.0.0.0
```

```
Router1(config-rip)#network 192.168.5.0
```

```
Router1(config-rip)#network 192.168.6.0
```

//Router2 routing protocol RIP

```
Router2(config)#router rip
```

```
Router2(config-rip)#network 2.0.0.0
```

```
Router2(config-rip)#network 6.0.0.0
```

```
Router2(config-rip)#network 7.0.0.0
```

//Router3 routing protocol RIP

```
Router3(config-rip)#network 3.0.0.0
```

```
Router3(config-rip)#network 7.0.0.0
```

```
Router3(config-rip)#network 8.0.0.0
```

//Router4 routing protocol RIP

```
Router4(config)#router rip
```

```
Router4(config)# network 4.0.0.0
```

```
Router4(config)# network 8.0.0.0
```

DHCP Services in All Block Using Server

Dhcp pool Add in Server And On The Service

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 4.0.0.10

DNS Server: 0.0.0.0

Start IP Address: 4.0.0.1

Subnet Mask: 255.0.0.0

Maximum Number of Users: 512

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverpool3	1.0.0.10	0.0.0.0	1.0.0.1	255.0.0.0	512	0.0.0.0	0.0.0.0
serverPool2	2.0.0.10	0.0.0.0	2.0.0.1	255.0.0.0	512	0.0.0.0	0.0.0.0
serverPool1	3.0.0.10	0.0.0.0	3.0.0.1	255.0.0.0	512	0.0.0.0	0.0.0.0
serverPool	4.0.0.10	0.0.0.0	4.0.0.1	255.0.0.0	512	0.0.0.0	0.0.0.0

☐ Top

Dhcp Relay Agent

Router>en

Router#config t

Router(config)#int fa0/0

Router(config-if)#ip helper-address 13.0.0.100

(NB : Same on all Router)

Go to the PC and Enable dhcp and Check

The screenshot shows the configuration window for PC8, specifically the Desktop tab. The IP Configuration section is active, showing settings for the FastEthernet0 interface. The DHCP option is selected, and the configuration fields are populated with values. A message indicates that the DHCP request was successful. The IPv6 Configuration section is also visible, with the Static option selected. The 802.1X section is partially visible at the bottom.

PC8

Physical Config **Desktop** Programming Attributes

IP Configuration [X]

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address: 1.0.0.1

Subnet Mask: 255.0.0.0

Default Gateway: 1.0.0.10

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::206:2AFF:FEFE:1AD0

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

AAA

AAA in Server

The screenshot shows the 'Server0' configuration window with the 'Services' tab selected. The 'AAA' service is highlighted in the left sidebar. The main configuration area is titled 'AAA' and includes the following sections:

- Service:** A toggle switch is set to 'On'. The 'Radius Port' is set to '1645'.
- Network Configuration:**
 - Client Name: 'E', Client IP: '4.0.0.10', Secret: '1234', ServerType: 'Radius'.
 - A table with columns 'Client Name', 'Client IP', 'Server Type', and 'Key'. It contains one entry: '1 E 4.0.0.10 Radius 1234'.
 - Buttons: 'Add', 'Save', 'Remove'.
- User Setup:**
 - Username: [empty], Password: [empty].
 - A table with columns 'Username' and 'Password'. It contains three entries: '1 bijo 5678', '2 bijoy 1234', and '3 rakesh 1234'.
 - Buttons: 'Add', 'Save', 'Remove'.

E(config-line)#aaa new-model

E(config)#username new secret newl

E(config)#aaa authentication login default group radius local

E(config)#aaa authentication enable default group radius

local E(config)#radius-server host 4.0.0.100 key 1234

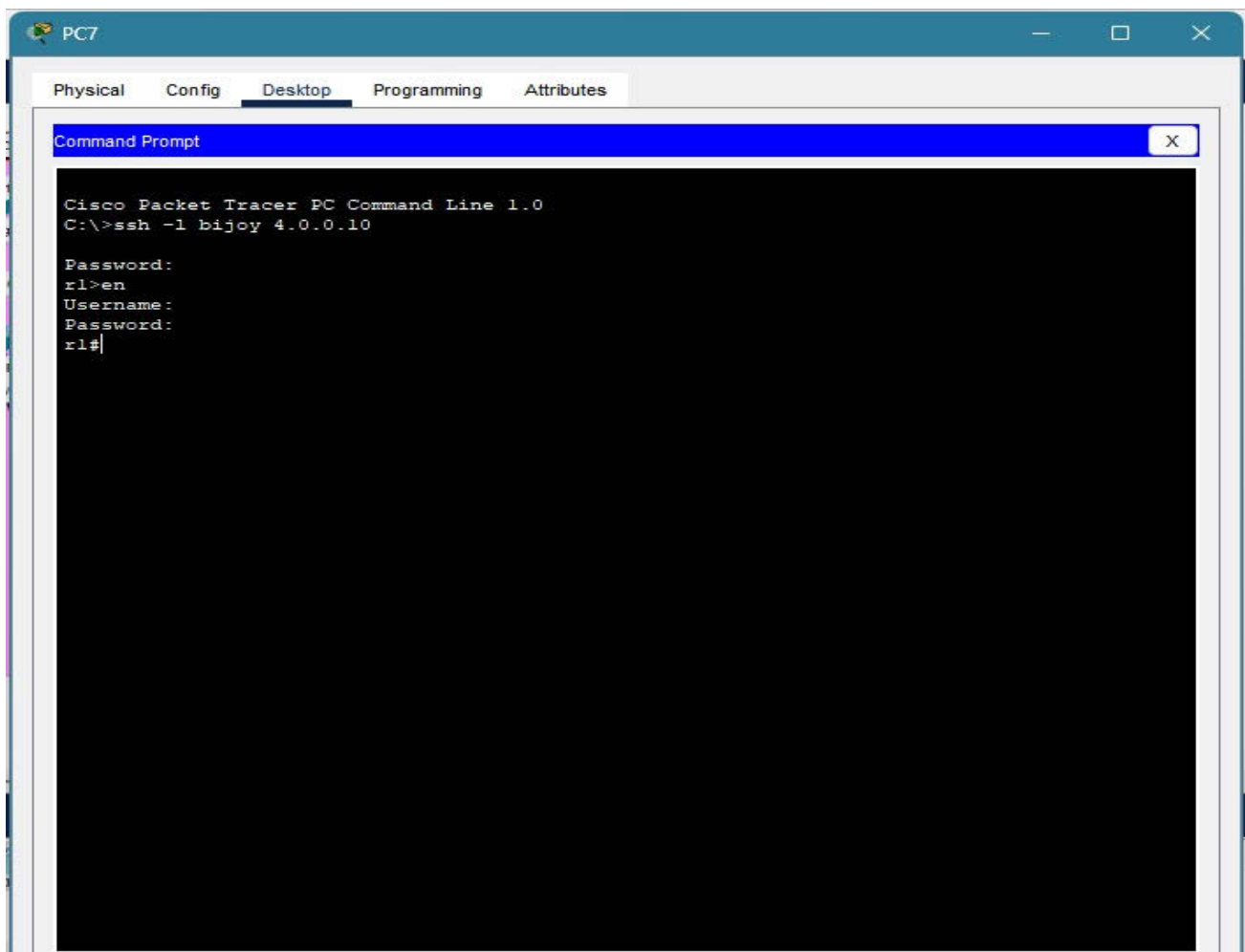
E(config)#ip domain-name cisco.com

E(config)#crypto key generate rsa

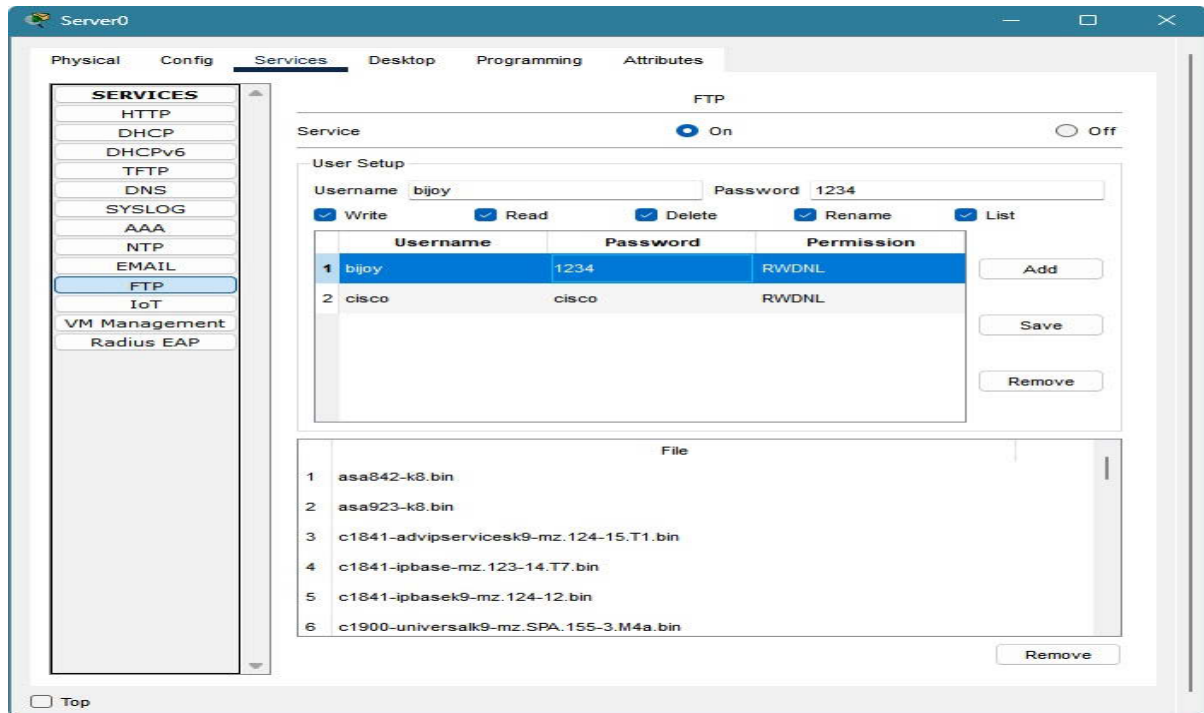
E(config)#line vty 0 4

E(config-line)#transport input ssh

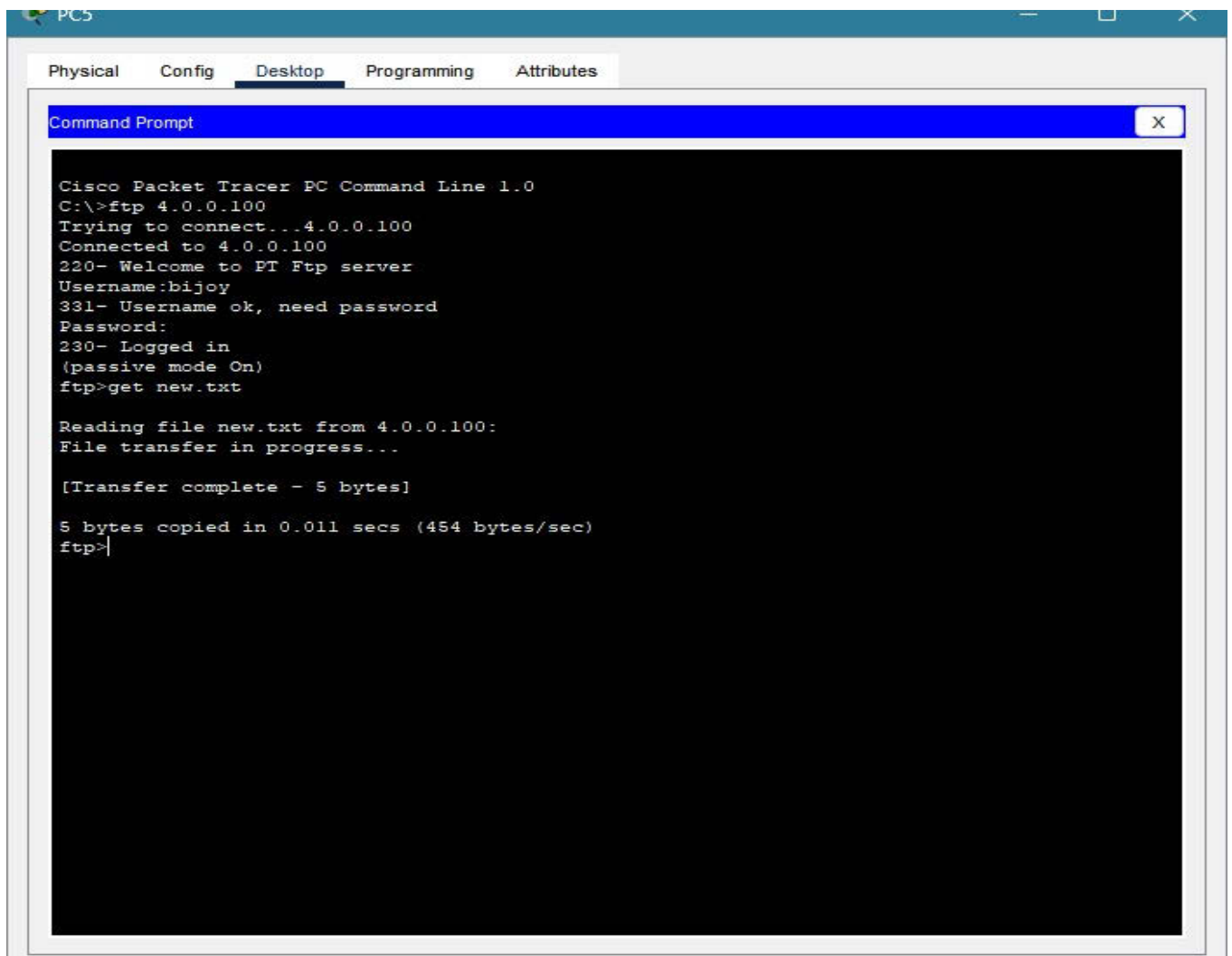
E(config-line)#login authentication default



Enable FTP



Check any PC



HSRP in Block E

Router>

Router>en

Router#config t

Enter configuration commands, one per line. End with CNTL/

Z. Router(config)#int gig0/0

Router(config-if)#standby 1 ip 13.0.0.201

Router(config-if)#standby 1 priority 111

*HSRP-6-STATECHANGE: Gigabyte Ethernet0/0 Grp 1 state
Standby ->LActive

Router(config-if)#standby 1 preempt

HSRP In New Router

Router#config t

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#int gig0/0

Router(config-if)#standby 1 ip 192.168.1.201

Router(config-if)#standby 1 ip 13.0.0.201

Router(config-if)#standby 1 priority 110

Router(config-if)#standby 1 preempt

*HSRP-6-STATECHANGE: Gigabyte Ethernet0/0 Grp 1 state

Speak ->LStandby

HSRP-6-STATECHANGE: Gigabyte Ethernet0/0 Grp 1 state
Standby -> Active

*HSRP-6-STATECHANGE: Gigabyte Ethernet0/0 Grp 1 state
Speak ->LStandby

ACL in Block D

Router>

Router>en

Router#config t

Enter configuration commands, one per line. End with CNTL/

Z. Router(config)#int gig0/0

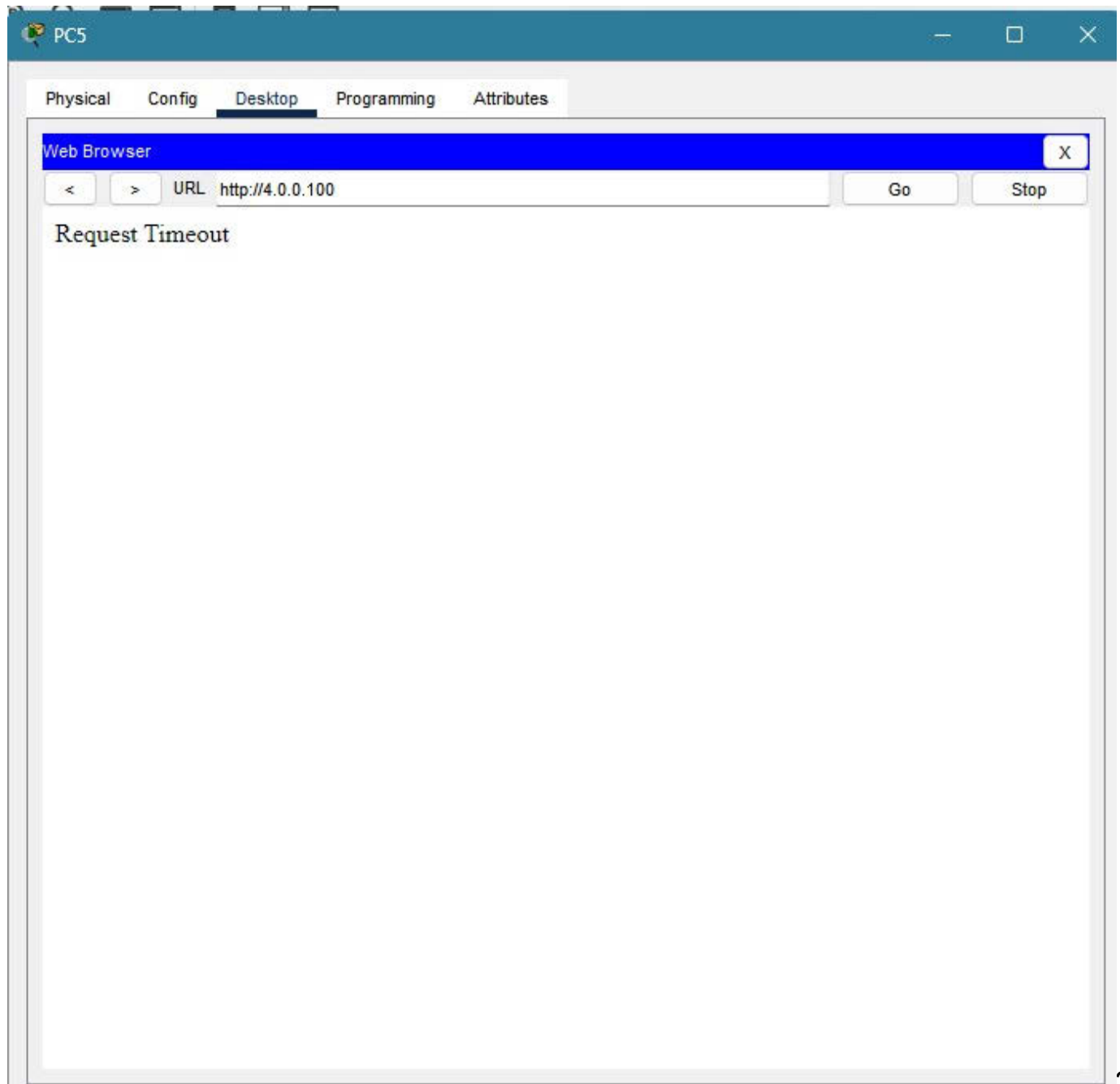
Router(config)#access-list 10 deny 3.0.0.2

Router(config)#access-list 10 permit any

Router(config)#int gig0/0

Router(config-if)#ip access-group 10 in

Check the any PC



Wireless Network in Block D

Setup In wireless Router

The screenshot shows a window titled "Wireless Router0" with three tabs: "Physical", "Config", and "GUI". The "Config" tab is active, and within it, the "Wireless" sub-tab is selected. On the left, a sidebar menu lists "GLOBAL" (with "Settings" and "Algorithm Settings" sub-items) and "INTERFACE" (with "Internet", "LAN", and "Wireless" sub-items). The main area displays the "Wireless Settings" configuration form. The "SSID" field contains "Dblock". The "Channel" dropdown is set to "6". Under the "Authentication" section, the "WEP" radio button is selected, and the "WEP Key" field contains "1234567890". Other options like "WPA-PSK", "WPA2-PSK", "WPA", and "WPA2" are unselected. The "RADIUS Server Settings" section includes fields for "IP Address" and "Shared Secret", both of which are empty. The "Encryption Type" dropdown is set to "40/64-Bits (10 Hex digits)".

Wireless Settings	
SSID	Dblock
Channel	6
Authentication	
☐ Disabled	☒ WEP
WEP Key: 1234567890	
☐ WPA-PSK	☐ WPA2-PSK
PSK Pass Phrase:	
☐ WPA	☐ WPA2
RADIUS Server Settings	
IP Address	
Shared Secret	
Encryption Type	40/64-Bits (10 Hex digits)

Setup In wireless Router

The image shows a configuration window titled "Smartphone0" with tabs for Physical, Config, Desktop, and Custom Interface. The Config tab is active, showing a left sidebar with "GLOBAL" (Settings, Algorithm Settings) and "INTERFACE" (Wireless0, 3G/4G Cell1). The main area displays the "Wireless0" configuration. The Port Status is checked and set to "On". Bandwidth is 300 Mbps. MAC Address is 0001.C7BB.8473. SSID is Dblock. Authentication is set to WEP with a key of 1234567890. Encryption Type is 40/64-Bits (10 Hex digits). IP Configuration is set to DHCP with IP Address 192.168.0.102 and Subnet Mask 255.255.255.0. IPv6 Configuration is set to Static.

Wireless0	
Port Status	☒ On
Bandwidth	300 Mbps
MAC Address	0001.C7BB.8473
SSID	Dblock
Authentication	
☐ Disabled ☒ WEP	WEP Key: 1234567890
☐ WPA-PSK ☐ WPA2-PSK	PSK Pass Phrase:
☐ WPA ☐ WPA2	User ID:
	Password:
Encryption Type	40/64-Bits (10 Hex digits)
IP Configuration	
☒ DHCP ☐ Static	
IP Address	192.168.0.102
Subnet Mask	255.255.255.0
IPv6 Configuration	
☐ DHCP ☐ Auto Config ☒ Static	

RESULT:

- Campus Network Management system is successfully obtained.
- By using Vlan10 and vlan 20, Admin and finance department are seperated in building A.
- Applying Extended Acl denied access of administrative block with building C,D and E.
- In building B by using vlan 30 and vlan 40, staff and HOD department are seperated respectively.
- Intervlan is applied to develop communication between vlans.
- Vtp is successfully used in Building D.
- Routing protocol RIP(Routing informative protocol) is appiled successfully.

